

The Job Sequencing and Tool Switching Problem

Part V: JGP + GSP Heuristic — Gap Analysis and Bounds

Scope and Status. This document analyses the *JGP + GSP decomposition heuristic*: solve the Job Grouping Problem (JGP) to get groups, then for each group select a configuration and solve a Group Sequencing Problem (GSP/GTSP) to find the best ordering and configurations jointly. We establish: (i) the JGP formulation and its relationship to the SSP objective; (ii) a verified counterexample showing the gap is non-zero; (iii) classical bounds ($Z_{\text{JGP}}^* \leq Z_{\text{SSP}}^* + 1$ and $Z_{\text{SSP}}^* \leq b \cdot Z_{\text{JGP}}^*$); (iv) a construction showing the gap grows without bound in K^* . **Open problem:** a tight bound for the gap when both K^* and b are fixed. This is marked explicitly as **TODO**.

Contents

1	The JGP + GSP Heuristic	2
1.1	The Job Grouping Problem (JGP)	2
1.2	From Groups to Configurations	2
1.3	The Group Sequencing Problem (GSP)	2
1.4	Lookahead Configurations Achieve the GSP Optimum	3
2	JGP vs. SSP: Fundamental Relationship	3
3	A Verified Counterexample: SSP Optimal $\neq$ JGP Optimal Configurations	4
4	Gap Growth: Constructing Unboundedly Large Differences	5
4.1	Copy-Paste Construction	5
4.2	Structure of the Gap: Two Sources	6
4.3	Mandatory Exit Lower Bound (Isolating Source 1)	6
5	Open Problems and TODO	7

The JGP + GSP Heuristic

The Job Grouping Problem (JGP)

Definition 1.1 (Feasible group). A subset of jobs $G \subseteq J$ is *feasible* if all their tools fit simultaneously into the magazine: $|\bigcup_{j \in G} T_j| \leq b$.

The JGP asks for the minimum number of feasible groups needed to partition J :

Definition 1.2 (JGP as a set covering problem [?]). Let $\mathcal{S}$ be the family of all feasible groups. Introduce binary variable $y_G \in \{0, 1\}$ (1 if group G is used).

$$\min \sum_{G \in \mathcal{S}} y_G \quad (1)$$

$$\begin{aligned} \text{s.t.} \quad & \sum_{G \in \mathcal{S}: j \in G} y_G \geq 1 \quad \forall j \in J \\ & y_G \in \{0, 1\} \quad \forall G \in \mathcal{S} \end{aligned} \quad (2)$$

The optimal value Z_{JGP}^* is the minimum number of groups (“batches”) needed.

Remark 1.3 (JGP hardness). The JGP is NP-hard [?] (it generalises graph colouring). Practical solution uses column generation: the pricing subproblem generates maximal feasible groups, equivalent to a bin-packing variant. Felipe Otiai’s thesis [?] develops an exact Branch-and-Price solver for the JGP.

From Groups to Configurations

Once the JGP yields groups $G_1, \dots, G_K$ (where $K \geq Z_{\text{JGP}}^*$), each group must be equipped with a *configuration* C_k : a maximal magazine state that covers all the group’s requirements.

Definition 1.4 (Configuration cluster for a group). For group G , the set of all valid configurations is:

$$\mathcal{C}(G) = \left\{ C \subseteq T \mid |C| = b, \bigcup_{j \in G} T_j \subseteq C \right\}.$$

The slack of group G is $s(G) = b - |\bigcup_{j \in G} T_j| \geq 0$; this is exactly the number of “free” padding slots.

The padding slots in C_k are the key degree of freedom: they can be filled with tools from *adjacent groups* to reduce the inter-group switching cost. This padding choice is central to the gap analysis.

The Group Sequencing Problem (GSP)

Given K groups and their configuration clusters, the *Group Sequencing Problem (GSP)* jointly selects one configuration per group and finds the optimal ordering of the groups to minimise total switching cost.

Formally, given groups $G_1, \dots, G_K$, choose $C_k \in \mathcal{C}(G_k)$ for each k and a permutation π of $\{1, \dots, K\}$ to minimise:

$$H(\pi, \{C_k\}) = \sum_{i=1}^{K-1} d(C_{\pi(i)}, C_{\pi(i+1)}) = \sum_{i=1}^{K-1} |C_{\pi(i+1)} \setminus C_{\pi(i)}|.$$

This is a **Generalised TSP (GTSP)** on clusters $\{\mathcal{C}(G_k)\}$ with arc costs inherited from the SSP metric d . For a fixed ordering π , the optimal configuration selection can be found greedily (lookahead padding, section 1.4). Over all orderings, the problem is a GTSP and is solved by standard GTSP/TSP solvers (Concorde, LKH-3, Noon-Bean transformation).

Definition 1.5 (JGP + GSP heuristic). Given a U-SSP instance:

1. Solve the JGP (exactly or approximately) to obtain groups $G_1, \dots, G_K$.
2. Solve the GSP (GTSP) over clusters $\{\mathcal{C}(G_k)\}$ to find the optimal permutation π^* and configurations $\{C_k^*\}$.
3. The resulting switching cost $H = H(\pi^*, \{C_k^*\})$ is a valid upper bound for the SSP: $H \geq Z_{\text{SSP}}^*$.

Important distinction. The GSP optimises over *all valid configurations* for each group, not just one fixed “canonical” choice. Any analysis that fixes the padding to a single strategy (e.g., all padding tools disjoint from adjacent groups) is studying a specific sub-optimal variant of the heuristic, not the JGP+GSP heuristic as defined above.

Lookahead Configurations Achieve the GSP Optimum

For a *fixed* ordering π of the groups, the optimal configuration for group $G_{\pi(k)}$ is obtained by *lookahead*: fill the $s(G_{\pi(k)})$ padding slots with as many tools from the next group $G_{\pi(k+1)}$ as possible.

Lemma 1.6 (Achievable per-transition cost). *For fixed adjacent groups G_k, G_{k+1} (in that order), the minimum switching cost achievable by any choice of configurations is:*

$$w^*(G_k, G_{k+1}) = \max\left(0, \left|\bigcup_{j \in G_k} T_j \setminus \bigcup_{j \in G_{k+1}} T_j\right| - s(G_{k+1})\right) = \max\left(0, \left|\bigcup_{j \in G_k} T_j \cup \bigcup_{j \in G_{k+1}} T_j\right| - b\right).$$

This lower bound is achieved by the lookahead configuration: $C_k^ = \left(\bigcup_{j \in G_k} T_j\right) \cup X_k$ where $X_k \subseteq \left(\bigcup_{j \in G_{k+1}} T_j\right) \setminus \left(\bigcup_{j \in G_k} T_j\right)$ and $|X_k| = \min(s(G_k), \left|\left(\bigcup_{j \in G_{k+1}} T_j\right) \setminus \left(\bigcup_{j \in G_k} T_j\right)\right|)$.*

Proof. Denote $T^{(k)} = \bigcup_{j \in G_k} T_j$. Configuration C_{k+1} must contain $T^{(k+1)}$ (mandatory, $|T^{(k+1)}|$ slots) and has $s_{k+1} = b - |T^{(k+1)}|$ free slots. Of the $|T^{(k)} \setminus T^{(k+1)}|$ tools in C_k not needed by G_{k+1} , at most s_{k+1} can be retained in C_{k+1} as padding. The remaining $\max(0, |T^{(k)} \setminus T^{(k+1)}| - s_{k+1})$ tools must leave, each costing one switch. The second equality follows from $|T^{(k)} \setminus T^{(k+1)}| - s_{k+1} = |T^{(k)} \cup T^{(k+1)}| - b$. The lookahead configuration pre-loads as many next-group tools as possible in the padding slots of C_k , maximising the tools that survive into C_{k+1} . $\square$ $\square$

JGP vs. SSP: Fundamental Relationship

The JGP optimal value Z_{JGP}^* and the SSP optimal value Z_{SSP}^* are related by two classical bounding theorems.

Theorem 2.1 (Lower bound: $Z_{\text{JGP}}^* - 1 \leq Z_{\text{SSP}}^*$ [?]). *The optimal SSP cost is at least $Z_{\text{JGP}}^* - 1$.*

Proof. Let σ^* be the optimal SSP sequence, traversing m distinct maximal configurations $W_1, W_2, \dots, W_m$ (in order of first use). Since this sequence processes all jobs and each W_r is a valid configuration, $\{W_1, \dots, W_m\}$ is a valid cover of J , so $m \geq Z_{\text{JGP}}^*$. Since consecutive distinct configurations differ in at least one tool, each of the $m - 1$ transitions costs ≥ 1

switch: $Z_{\text{SSP}}^* \geq m - 1 \geq Z_{\text{JGP}}^* - 1$. □ □

Theorem 2.2 (Upper bound: $Z_{\text{SSP}}^* \leq b \cdot Z_{\text{JGP}}^*$ [?]). *The optimal SSP cost is at most b times the optimal JGP value.*

Proof. Take any JGP-optimal partition into $K^* = Z_{\text{JGP}}^*$ groups $G_1, \dots, G_{K^*}$. Process jobs batch by batch. The worst-case transition between any two consecutive groups occurs when their configurations are completely disjoint, requiring b tool insertions. With $K^* - 1$ transitions and an initial load of $\leq b$ tools, total switches $\leq b + b(K^* - 1) = b \cdot K^* = b \cdot Z_{\text{JGP}}^*$. Since Z_{SSP}^* is the minimum, $Z_{\text{SSP}}^* \leq b \cdot Z_{\text{JGP}}^*$. □ □

Remark 2.3 (Are the bounds tight?). The lower bound $Z_{\text{JGP}}^* - 1 \leq Z_{\text{SSP}}^*$ is tight (e.g., any instance where the SSP processes each group as a single configuration with zero intra-group switches achieves $Z_{\text{SSP}}^* = Z_{\text{JGP}}^* - 1$). The upper bound $Z_{\text{SSP}}^* \leq b \cdot Z_{\text{JGP}}^*$ is extremely loose for structured instances; improving it is a primary research goal.

A Verified Counterexample: SSP Optimal $\neq$ JGP Optimal Configurations

The following example demonstrates that (a) the configurations used in an optimal SSP solution need not come from a JGP-optimal grouping, and (b) even with the best JGP-optimal grouping, the JGP+GSP heuristic may not achieve Z_{SSP}^* .

Example 3.1 (6-job ring instance). Let $T = \{t_1, \dots, t_6\}$, $b = 3$, and six jobs with requirements:

$$T_{j_1} = \{t_1, t_2\}, \quad T_{j_2} = \{t_2, t_3\}, \quad T_{j_3} = \{t_3, t_4\}, \quad T_{j_4} = \{t_4, t_5\}, \quad T_{j_5} = \{t_5, t_6\}, \quad T_{j_6} = \{t_6, t_1\}.$$

This is a “ring” topology: consecutive jobs share exactly one tool. The conflict graph has $|T_{j_i} \cup T_{j_k}| = 3 = b$ for adjacent pairs and $|T_{j_i} \cup T_{j_k}| \geq 3$ for non-adjacent pairs, forcing every pair of jobs needing disjoint tools into different groups.

JGP optimal solution ($Z_{\text{JGP}}^* = 3$). The minimum number of feasible groups is 3 (verified by PORTA enumeration). Two optimal groupings exist; one example:

$$G_1 = \{j_1, j_2\}, \quad G_2 = \{j_3, j_4\}, \quad G_3 = \{j_5, j_6\}.$$

Mandatory tool sets: $T^{(1)} = \{t_1, t_2, t_3\}$, $T^{(2)} = \{t_3, t_4, t_5\}$, $T^{(3)} = \{t_1, t_5, t_6\}$. Each has slack $s = 0$, so configurations are unique: $C_1 = \{t_1, t_2, t_3\}$, $C_2 = \{t_3, t_4, t_5\}$, $C_3 = \{t_1, t_5, t_6\}$.

JGP+GSP result. With zero slack, there is only one configuration per group; the GSP just finds the best ordering. In any ordering, each pair of adjacent configurations (under the cyclic ring structure) shares exactly one tool, giving $d = b - 1 = 2$ per transition. The best ordering has 2 transitions: total cost $= 2 \times 2 = 4$.

SSP optimal solution ($Z_{\text{SSP}}^* = 3$). Sequence $j_1 \rightarrow j_2 \rightarrow j_3 \rightarrow j_4 \rightarrow j_5 \rightarrow j_6$, processed continuously via KTNS with $b = 3$:

Step	Job	Magazine	Switches
1	j_1	$\{t_1, t_2, t_3\}$	3 (initial load)
2	j_2	$\{t_1, t_2, t_3\}$	0 (no change)
3	j_3	$\{t_1, t_3, t_4\}$	1 ($-t_2, +t_4$; t_1 retained)
4	j_4	$\{t_1, t_4, t_5\}$	1 ($-t_3, +t_5$; t_1 retained)
5	j_5	$\{t_1, t_5, t_6\}$	1 ($-t_4, +t_6$; t_1 retained)
6	j_6	$\{t_1, t_5, t_6\}$	0 (no change)

Total (excluding initial load): $Z_{\text{SSP}}^* = 3$. KTNS retains t_1 as a “bridge” tool across jobs j_2 – j_5 , smoothly rolling the ring. The JGP with 3 groups severs this bridge at both group boundaries.

Gap. JGP+GSP gives $H = 4$, SSP optimum is $Z_{\text{SSP}}^* = 3$. **Gap** = 1.

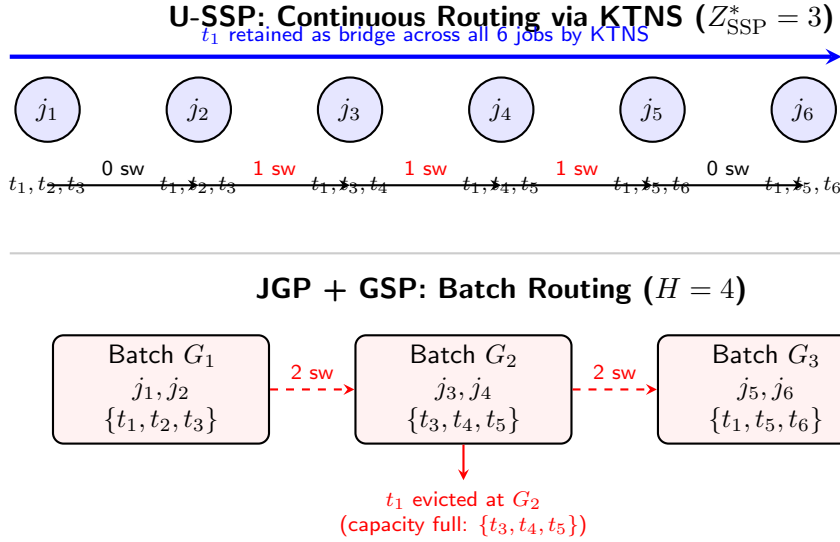


Figure 1: Six-job ring instance ($b = 3$). KTNS retains tool t_1 continuously across all jobs (3 switches total). The JGP must use 3 batches; each batch saturates the magazine with its mandatory tools, forcing t_1 to be evicted at batch G_2 and reloaded at G_3 (4 switches total). Gap = 1.

Gap Growth: Constructing Unboundedly Large Differences

The 6-job ring example achieves a gap of 1 with $K^* = 3$ groups. We now show the gap can be made as large as desired.

Copy-Paste Construction

Theorem 4.1 (Gap grows with K^*). *For any integer $g \geq 1$, there exists a U-SSP instance with $Z_{\text{JGP}}^* = 3g$ groups where the JGP+GSP heuristic costs at least g more switches than Z_{SSP}^* .*

Proof. Take g independent copies of the 6-job ring instance from Example 3.1, using completely disjoint tool sets (e.g., copy ℓ uses tools $\{t_{6\ell-5}, \dots, t_{6\ell}\}$). Process all copies in sequence.

Each copy independently has $Z_{\text{JGP}}^{*(\ell)} = 3$ and $Z_{\text{SSP}}^{*(\ell)} = 3$, with gap 1. Since the copies share no tools, configurations from different copies are fully disjoint and the inter-copy transitions cost exactly b switches regardless.

JGP+GSP: The g copies produce $3g$ groups total. Within each copy, the GSP pays 4 switches (the per-copy gap is 1). Across copies, transitions cost $b = 3$ each (fully disjoint configurations), giving $g - 1$ inter-copy transitions of cost 3. Total: $H = 4g + 3(g - 1)$.

SSP optimum: $Z_{\text{SSP}}^* = 3g + 3(g - 1)$ (3 switches per copy, $b = 3$ between copies).

Gap: $H - Z_{\text{SSP}}^* = (4g + 3(g - 1)) - (3g + 3(g - 1)) = g$.

Since g is arbitrary, the gap is unbounded. $\square$ $\square$

Remark 4.2. In this construction, $K^* = 3g$ grows with g . The gap is $g = K^*/3$. The gap grows *proportionally* to K^* , suggesting the relevant quantity for bounding the gap is K^* itself, not just b . This aligns with the loose upper bound: $Z_{\text{SSP}}^* \leq b \cdot K^*$, so the gap $H - Z_{\text{SSP}}^* \leq H \leq b \cdot K^*$.

Structure of the Gap: Two Sources

The gap between H (JGP+GSP cost) and Z_{SSP}^* has two independent sources:

1. **Padding-choice gap:** Even with the JGP-optimal partition into K^* groups, the GSP may not achieve Z_{SSP}^* because the best configurations for those groups (with lookahead padding) may still cost more than the SSP optimal (which can choose any sequence, including ones that dissolve group boundaries).
2. **Grouping-choice gap:** The SSP optimal may use more than K^* groups (e.g., $K^* + 1$ groups with smaller mandatory sets), allowing configurations with more slack and hence cheaper transitions.

In the 6-job ring, source (2) is zero (using 4 groups does not help), and source (1) accounts for the full gap of 1. In the copy-paste construction, both sources contribute.

Mandatory Exit Lower Bound (Isolating Source 1)

For instances where the grouping-choice gap (source 2) is zero, the following lower bound on Z_{SSP}^* is tight for the specific instance family.

Definition 4.3 (Mandatory exit tools). For a JGP-optimal grouping $\{G_1, \dots, G_{K^*}\}$ ordered as $G_1 \rightarrow \dots \rightarrow G_{K^*}$, define the *mandatory exit set* at boundary k as:

$$D_k = \left(\bigcup_{j \in G_k} T_j \right) \setminus \left(\bigcup_{i > k} \bigcup_{j \in G_i} T_j \right), \quad k = 1, \dots, K^* - 1.$$

These are tools required by G_k that are never needed again in any later group.

Lemma 4.4 (Mandatory exit lower bound). *For any feasible SSP solution (any grouping, any ordering), the total number of tool switches is at least $\sum_{k=1}^{K^*-1} |D_k|$.*

Proof. Fix any tool $t \in D_k$. By definition, t is required by some job $j \in G_k$ and is never required by any job in $G_{k+1}, \dots, G_{K^*}$.

- t must be in the magazine when j is processed (either from the initial load or via an insertion before j).
- After the last job requiring t is processed, t will eventually be evicted (since subsequent jobs require tools not currently loaded, and the magazine has finite capacity b).
- This eviction corresponds to at least one *removal event*. In the uniform SSP model, $|C_u| = |C_v| = b$ for all transitions, so removals = insertions per transition, and each removal event corresponds to exactly one switch.

Since the sets $D_1, \dots, D_{K^*-1}$ are pairwise disjoint (by the “never needed again” definition), the removal events are distinct, giving at least $\sum_k |D_k|$ total switches. $\square$ $\square$

Remark 4.5 (Application). For the sliding-window construction of `06_grouping_selection.tex`, $|D_k| = r - \text{ov}$ for all k (where r is the group tool set size and ov is the overlap between adjacent groups), giving a lower bound of $(K^* - 1)(r - \text{ov})$ on Z_{SSP}^* . Paired with the lookahead upper bound of the same value, this shows $Z_{\text{SSP}}^* = (K^* - 1)(r - \text{ov})$ for that family.

Open Problems and TODO

TODO 1: Tight bound for fixed K^* and b .

The copy-paste construction shows the gap $H - Z_{\text{SSP}}^*$ can be made proportional to K^* . The copy-paste construction uses g independent sub-instances, each with its own 3 optimal groups, so $K^* = 3g$. But it is an open question whether, for a *fixed* K^* (e.g., $K^* = 3$) and a *fixed* b , the gap can exceed 1.

Progress: Computational experiments via PORTA on small instances suggest the gap is at most 1 when $K^* = 3$, but no proof exists. The goal is to answer:

For fixed K^ and b , what is the maximum possible gap between the JGP+GSP heuristic (with optimal configurations and ordering) and Z_{SSP}^* ?*

A tight construction (instance achieving the gap) and a matching upper bound proof are both needed.

TODO 2: Effect of sub-optimal groupings on the gap.

The Theorem 4.1 construction uses JGP-optimal groupings. A sub-optimal grouping with $K > K^*$ groups has more slack per group, potentially enabling lookahead configurations with smaller transitions. It is conjectured that some sub-optimal groupings yield $H <$ the best JGP-optimal result. Proving (or disproving) this requires:

- A formal characterisation of which sub-optimal groupings are beneficial.
- A bound on how much extra slack is needed to close the gap.

See `06_grouping_selection.tex` for related directions.